

Week 10 Revision Selected Questions

5CM522 Operational Research

Reformatted from Sem 2 Week 10 - Revision.pdf

This document contains all selected questions from the Week 10 revision sheet, reformatted as a clean booklet. The questions have been renumbered so that the pack reads continuously.

1.

GreenGrid Energy is choosing a set of improvement projects for a regional depot network. Each project is either undertaken in full or not undertaken at all.

Project	Cost (£m)	Annual Benefit Score
<i>A</i>	6	14
<i>B</i>	8	17
<i>C</i>	5	11
<i>D</i>	9	20
<i>E</i>	4	9
<i>F</i>	7	15

The total budget is £21 million.

The following additional conditions apply:

- if project *D* is selected, project *B* must also be selected;
 - projects *A* and *F* cannot both be selected;
 - at least one of projects *C* and *E* must be selected.
- (a) Formulate a binary integer programming model to maximise total annual benefit subject to the budget and project-selection rules. Clearly define all decision variables. (8 Marks)
- (b) Solve the model in Python or by systematic analysis, and determine which projects should be selected and the maximum total annual benefit score. (7 Marks)
- (c) Explain why this problem requires binary decision variables rather than the continuous variables used in ordinary linear programming. (3 Marks)
- (d) Modify the model so that exactly four projects must be selected, then resolve or explain how the optimal solution would change. (4 Marks)
- (e) Suggest one realistic extension that would make this capital-selection model more useful in practice. No full formulation is required. (3 Marks)

Total 25 Marks

2.

Hock-Brooke (Automobiles), the main dealer in Ceeham, receives its new stock on Monday. Most customers collect their new cars on Friday or Saturday. New cars are stored initially in an old warehouse to be prepared for display in the showroom or, when the warehouse is full, in a secure parking area.

The dealership is about to replace its old large warehouse with a modern purpose-built warehouse and needs to know the most suitable size for this new warehouse.

The following data are available:

Number of Cars Delivered	7	8	9	10
Frequency (%)	10	20	50	20

Weekly Demand	6	7	8	9	10	11
Frequency (%)	5	20	30	10	20	15

The cost of providing a single warehouse space is £10 per week, and the cost of hiring an additional parking space is £20 per week.

- (a) Simulate the operation of this dealership with a warehouse capacity of 9 cars and hence determine the average cost of using the warehouse per week. (15 Marks)
- (b) Is 9 the best capacity? If not, determine the best possible capacity for the warehouse. (10 Marks)

Total 25 Marks

3.

A university library help desk is modelled as a queueing system. Students arrive randomly and service times are assumed to follow an exponential distribution. You should adapt the SimPy structures used in class, but tailor the model to this setting.

Use the following assumptions:

Quantity	Value
Simulation length	360 minutes
Average service time	7 minutes
Baseline arrival rate	10 students per hour
Peak arrival rate	15 students per hour
Service target	97% wait less than 8 minutes
Priority users	10% of arrivals
Suggested random seed	24

- (a) Describe the main elements of a discrete-event simulation model for this system: entities, resources, events, stochastic inputs and performance measures. (5 Marks)
- (b) Adapt a SimPy model to represent the baseline system with 2 advisers and an arrival rate of 10 students per hour. Report the total served, average wait, maximum wait and percentage waiting less than 8 minutes. (6 Marks)
- (c) Investigate the peak-period scenario with 2, 3 and 4 advisers. Present your results in a clearly labelled table and determine the minimum number of advisers needed to meet the service target. (4 Marks)
- (d) Extend the model so that 10% of users are priority users who should be served ahead of standard users. Explain how the model should be changed and discuss the likely effect on standard-user waits. (6 Marks)
- (e) Explain why a single run is not enough to support a staffing decision and describe how you would use replications and/or a warm-up period to improve the analysis. (4 Marks)

Total 25 Marks

4.

F.B. Knight (Deliveries) distributes goods from its warehouses to the WalShop supermarkets throughout the East Midlands. To optimise profits for this fixed-price contract, the company needs to determine its best delivery schedule from warehouse to shop.

- (a) Use the following data to determine F.B. Knight's optimal distribution plan.

Warehouse	<i>A</i>	<i>B</i>	<i>C</i>
Available Stock	100	100	100

Shop	<i>X</i>	<i>Y</i>	<i>Z</i>
Demand	70	80	150

The shipping costs per 10 items are:

	<i>X</i>	<i>Y</i>	<i>Z</i>
<i>A</i>	8	7	5
<i>B</i>	12	3	9
<i>C</i>	10	11	7

(10 Marks)

- (b) Determine how the solution changes if the following additional restrictions are imposed:

B cannot deliver to *Y*

and

A must deliver at least 5 items to *Y*.

(5 Marks)

- (c) Two new distribution locations act as intermediary nodes between the warehouses and shops and become available for use. Determine whether you should use one or both locations. Assume that if you use the locations, you do not ship directly from warehouses to shops.

Warehouse to intermediary costs per 10 items:

	<i>D</i> ₁	<i>D</i> ₂
<i>A</i>	4	5
<i>B</i>	3	2
<i>C</i>	6	4

Intermediary to shop costs per 10 items:

	<i>D</i> ₁	<i>D</i> ₂
<i>X</i>	4	2
<i>Y</i>	5	4
<i>Z</i>	2	4

(10 Marks)

Total 25 Marks

5.

The data analyst at N&F has started an initial analysis of sales data within the firm. The following table shows quarterly sales data over a 4-year period.

Period	Season	Sales Data
1	Spring	1013
2	Summer	1025
3	Autumn	1002
4	Winter	1021
5	Spring	1053
6	Summer	1067
7	Autumn	1042
8	Winter	1060
9	Spring	1093
10	Summer	1106
11	Autumn	1083
12	Winter	1101
13	Spring	1134
14	Summer	1147
15	Autumn	1123
16	Winter	1139

Use this data to compare and contrast the use of both additive and multiplicative models to forecast sales.

- (a) Plot the sales data and smoothed (trend) data. (2 Marks)
- (b) Calculate seasonal variations using both additive and multiplicative models for seasonal variation. (8 Marks)
- (c) Produce sales forecasts for next year using both models. (12 Marks)
- (d) Evaluate the results from these forecasting models and make a recommendation concerning the most appropriate model to be used to forecast future sales. (3 Marks)

Total 25 Marks

End of Selected Questions